Glossary:

MVP: minimal viable product
ARTA: shorthand for the risk mitigation strategies
PSI: potentially shippable increment
Feature: same as backlog item
Product backlog item: a planned feature
Sprint backlog item: a feature being worked on in a past or current sprint
DoD: definition of done: a measure of implementation quality has all the features previously agreed upon been implemented + tested + are they working without errors.
Skateboard: Proof of concept version demonstrating viability of a complete project.
Wheel: Proof of concept of a (small) part of the project, parts of which are thus unimplemented. This must be individually testable and runnable.
CR: code review
RA: risk assesment
RMP: Risk mitigation plan

Capstone Assignment 1

Submit this form as a MS Word docx file to Amathuba.
Also check it in to your Gitlab repo as a MS word docx file, so that team members, tutors, and clients have access to it.

Initial project plan [10% of total mark] *Do not edit this document after submission to Amathuba*			
Project name:	CommHub		
Team nickname			
Team names (first/last)	Dzanga	Faresa	Kholofelo
	Ravhutsi	Malitsha	Zondi
Student numbers	RVHDZA001	MLTFAR001	ZNDKHO003
Project description *1 or 2 lines*	Building a WebApp to serve as the community hub utilising iNethi technologies and deployed to serve Ocean View		
Planned programming language(s)	Python (backend), MySQL (database), HTML, CSS, JavaScript (frontend)		
sGit url	https://gitlab.cs.uct.ac.za/zndkho003/csc-capstone		
Sprint length	1 Week		
Describe non-backlog/non-coding work *(filling in forms is not 'work')*			
Work item 1	UX design		
Person responsible	Dzanga		
Add more non-backlog items as needed, you need as least as many of such items as there are team members. Only add 1 or 2 major items per team member.			
Work item 2	Database design		
Person responsible	Faresa		
Add more non-backlog items as needed, you need as least as many of such items as there are team members. Only add 1 or 2 major items per team member.			
Work item 3	iNethi Website deployment set-up		
Person responsible	Kholofelo		

Add more non-backlog items as needed, you need as least as many of such items as there are team members. Only add 1 or 2 major items per team member.			
Risk analysis			
Describe Risk 1	**Team member is unavailable due to illness, etc.** *do not remove or alter this risk.*		
Likelihood	Medium	Impact	High
ARTA strategy	Accept		
Mitigations	Use regular Git pushes to ensure all members stay up to date, have multiple knowledge sharing sessions and when necessary, remote scrum meetings		
Describe Risk 2	Web interface vulnerability through SQL injection		
Likelihood	Low	Impact	High
ARTA strategy	Reduce		
Mitigations	Prevent malicious dynamic SQL queries written with string concatenation from being executed. All queries must be parametrised, validated and sanitized		
Describe Risk 3	Database breach resulting in release and/or loss of sensitive private user information		
Likelihood	Low	Impact	High
ARTA strategy	Reduce		
Mitigations	Restrict database permissions to admins and use hashed passwords to protect against data leaks. Additionally, perform regularly scheduled backups of the database		
Describe Risk 4	Testing delayed by server or deployment issues		
Likelihood	Low	Impact	Medium
ARTA strategy	Reduce		
Mitigations	Maintain separate development and production environments while deploying prototypes early and keeping regular backups of configuration files		
Describe Risk 5	Essential requirements change after the development has started and the production backlog has been finalised		
Likelihood	Medium	Impact	Medium
ARTA strategy	Reduce		
Mitigations	Hold regular client meetings and finalise the product backlog early into development.		
Project backlog items			
Add as many backlog items as needed			
Backlog item 1			
Name	*Account Management*		
Description	*User wants to create/delete account to access community features or remove personal information when they no longer want to be part of the platform*		

Priority	*High*
Main developer	*Faresa*
Estimated time	*2 days*
Acceptance criteria (DoD):when user can register, login and delete their account	
Functionality	*User can register, log in, and permanently delete their account and data*
Test cases	1. *Valid signup succeeds and creates DB record with correct role* 2. *Duplicate/invalid signup is rejected* 3. *Deleted account can no longer log in and personal data is removed*
Client remarks	
Approved by client	*signature*
Backlog item 2	
Name	*Service directory*
Description	*Community member wants to post, search/browse, and manage applicants for services to offer or find help within the community*
Priority	*High*
Main developer	*Faresa*
Estimated time	*1 week*
Acceptance criteria (DoD):	
Functionality	*User can create a listing, search/browse existing listings, apply to one and view and accept/decline applicants (if owner)*
Test cases	1. *Listing saves and appears in search* 2. *Search/filter returns correct matches* 3. *Accept/decline updates for both parties*
Client remarks	
Approved by client	*signature*
Backlog item 3	
Name	*Research management*
Description	*Researcher wants to build, publish, close and analyse surveys, capture informed consent, and track honorarium payments to run research end-to-end*
Priority	*High*
Main developer	*Dzanga*
Estimated time	*2 weeks*
Acceptance criteria (DoD): if user can add and remove survey and see research project they are running	
Functionality	*Researcher can build a survey, requires consent before entry, publish/close it, opt in applicants, view aggregated results, and mark participants as paid*
Test cases	1. *Closed surveys removed from listings* 2. *Accept/decline applicants* 3. *Results match actual submissions* 4. *Payment status persists correctly* 5. *Survey can be posted*

	6. *Survey can be deleted*
Client remarks	
Approved by client	*signature*
Backlog item 4	
Name	*Admin and Moderation*
Description	*Admin wants to moderate services, manage users, and enforce role-based access to keep platform safe and appropriate*
Priority	*High*
Main developer	*Kholofelo*
Estimated time	*4 days*
Acceptance criteria (DoD):	
Functionality	*Admin can flag/remove listings, process user deletion requests, and system restricts pages/actions by role*
Test cases	1. *Removed listing disappears from directory* 2. *Deleted accounts can no longer log in* 3. *Non-admin cannot access admin routes*
Client remarks	
Approved by client	*signature*
Backlog item 5	
Name	*Platform hosting*
Description	*Deploy platform as a Python web app on the iNethi server for offline/local access, and make it accessible online from outside network*
Priority	*High*
Main developer	*Kholofelo*
Estimated time	*4 days*
Acceptance criteria (DoD):	
Functionality	*Platform runs and is usable via the local iNethi network with no internet and is also reachable via a public URL.*
Test cases	1. *Site loads with internet disabled on local network* 2. *Site loads from external network*
Client remarks	
Approved by client	*signature*
Backlog item 6	
Name	*Database setup*
Description	*Design and implement an SQL database to store all platform data*
Priority	*High*
Main developer	*Faresa*
Estimated time	*3days*
Acceptance criteria (DoD):	
Functionality	*Schema supports all core entities with correct relationships*
Test cases	1. *Data written by app is retrievable without loss* 2. *Foreign key constraints prevent orphaned records*

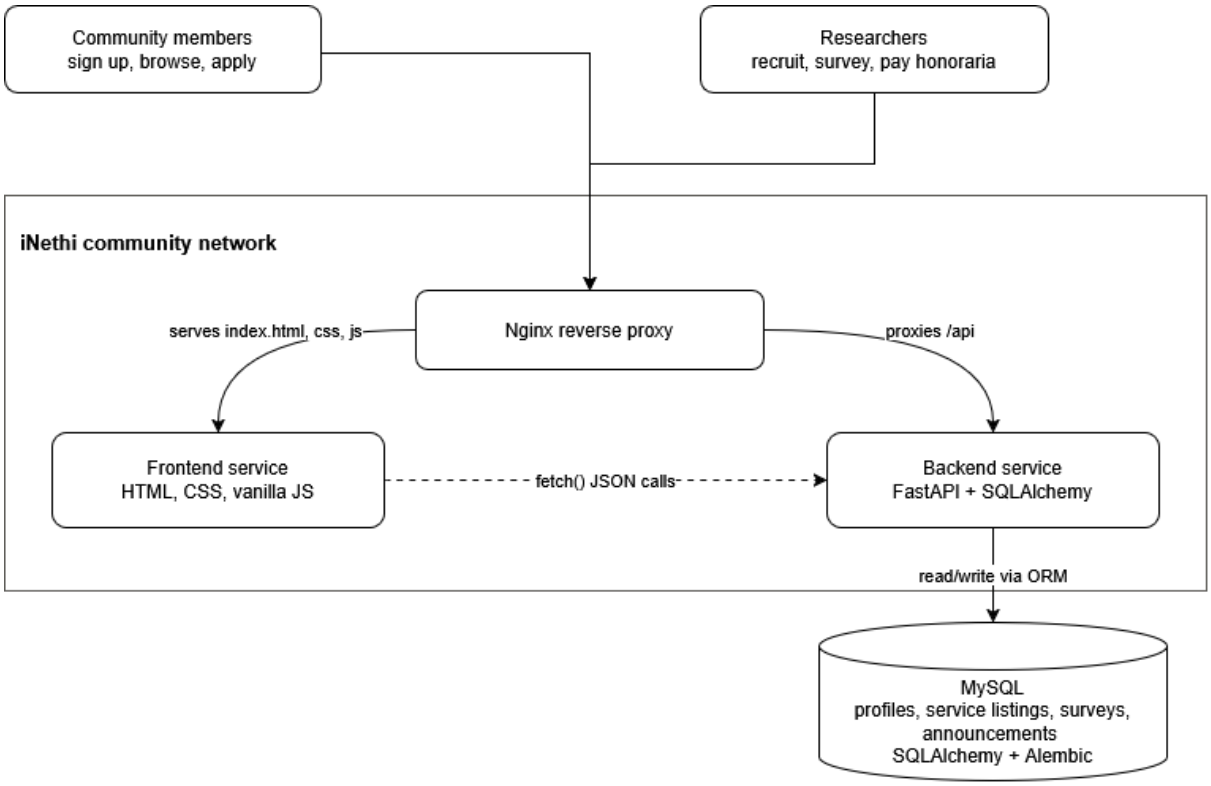
Client remarks	
Approved by client	*signature*
Backlog item 7	
Name	*Mobile Landing screens*
Description	*Design mobile landing screens so the platform is useable on cellphones and tablets*
Priority	*Middle*
Main developer	*Khoolfelo*
Estimated time	*4days*
Acceptance criteria (DoD):	
Functionality	*Landing screen renders correctly and is fully navigable*
Test cases	1. *No layout overflow on small screens* 2. *Key actions reachable with 2 taps on mobile* 3. *Consistent*
Client remarks	
Approved by client	*signature*
Backlog item 8	
Name	*Trust*
Description	*Community member wants to understand how data is used and stored to trust app with information*
Priority	*High*
Main developer	*Faresa*
Estimated time	*1 day*
Acceptance criteria (DoD):	
Functionality	*User can see how data is used so app adheres to POPIA*
Test cases	1. *App shows and explains how data is used*
Client remarks	
Approved by client	*signature*
Backlog item 9	
Name	*Community statistics*
Description	*Community member wants to view general statistics to understand community needs*
Priority	*Low*
Main developer	*Dzanga*
Estimated time	*3days*
Acceptance criteria (DoD):	
Functionality	*User can see general statistics*
Test cases	1. *App shows correct statistics from actual data*
Client remarks	
Approved by client	*signature*

Add more backlog items as needed.
A backlog item must always be completable in a single sprint.
Remember: these are all backlog items for the entire project.
During the project new items may be added or removed as needed, but only in consultation with the client!

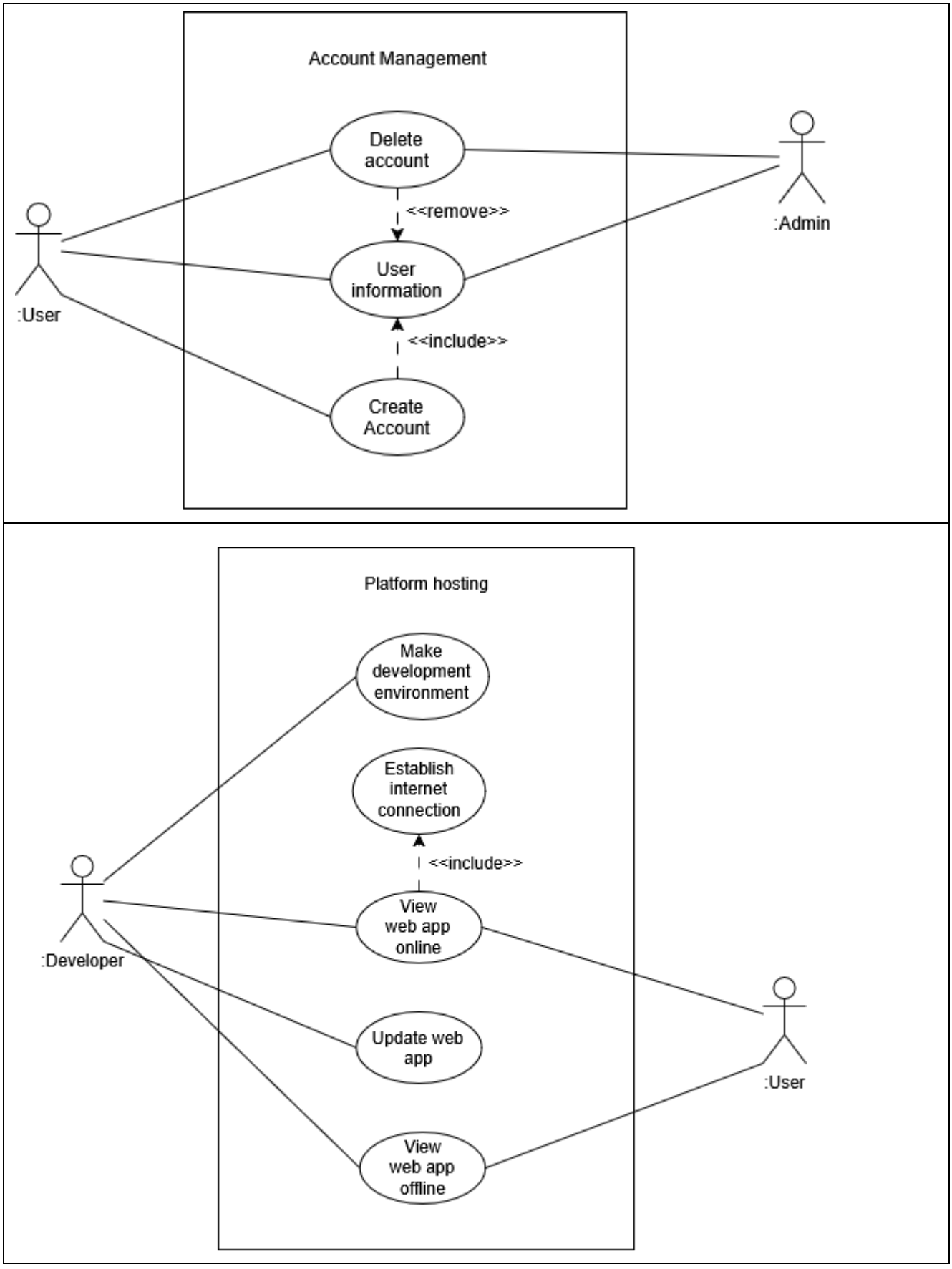
GenAI declaration

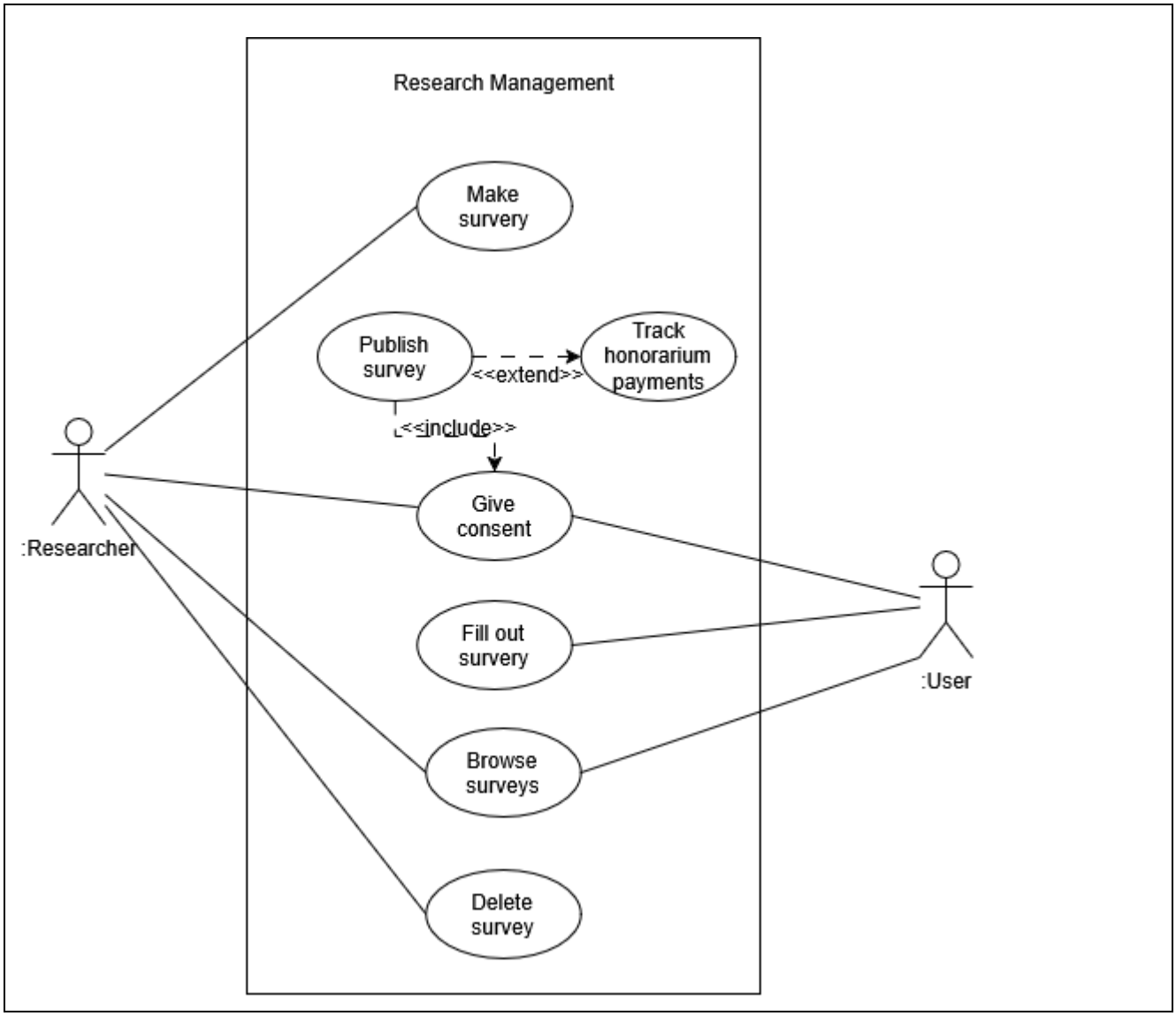
Did you use GenAI	*N*
Describe use	*Describe the nature and extent of GenAI use and the specific model used. You are allowed to use GenAI, but you **must** adhere to the GenAI policy shown on the CSC3003S Amathuba site.*

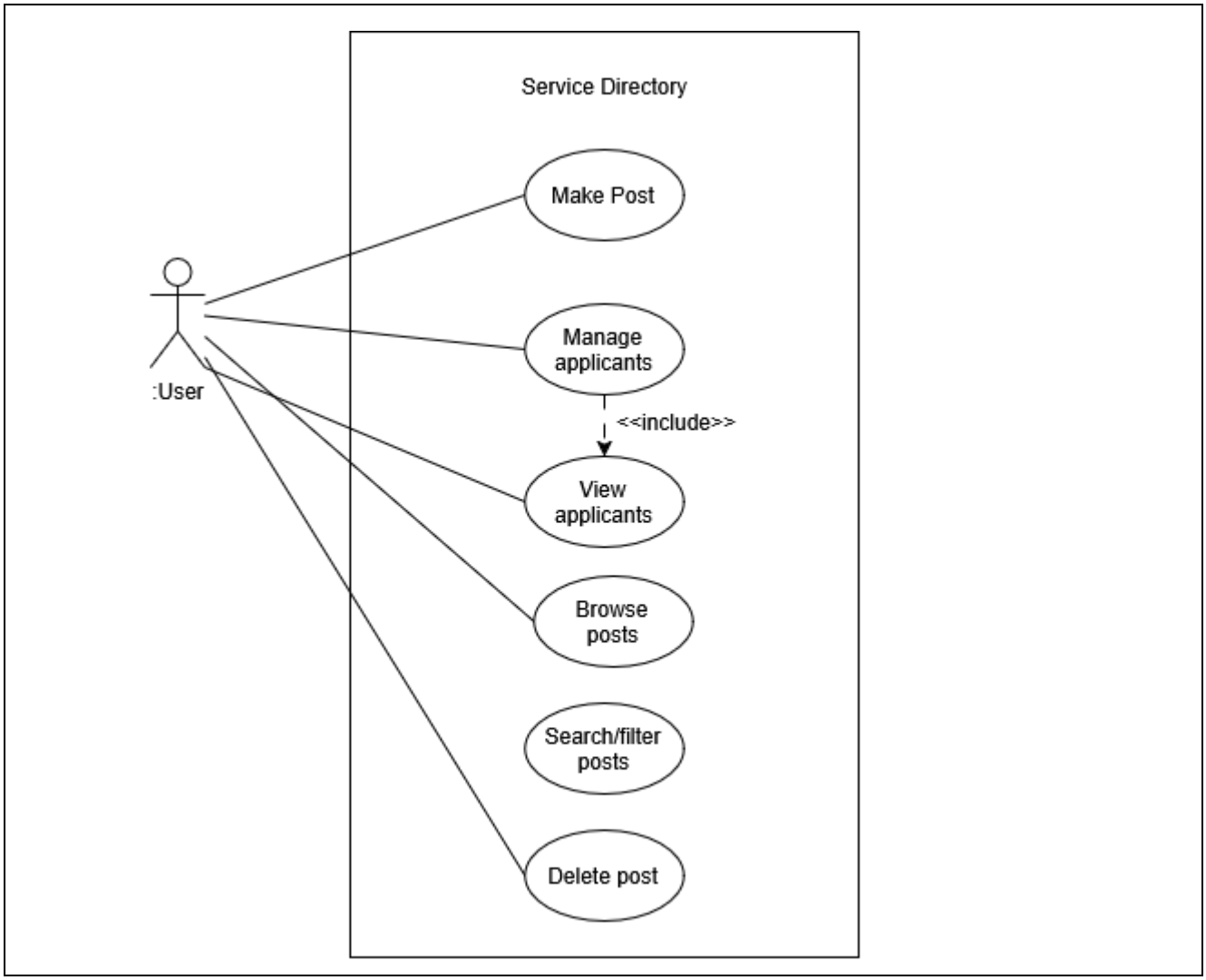
Architecture Diagram



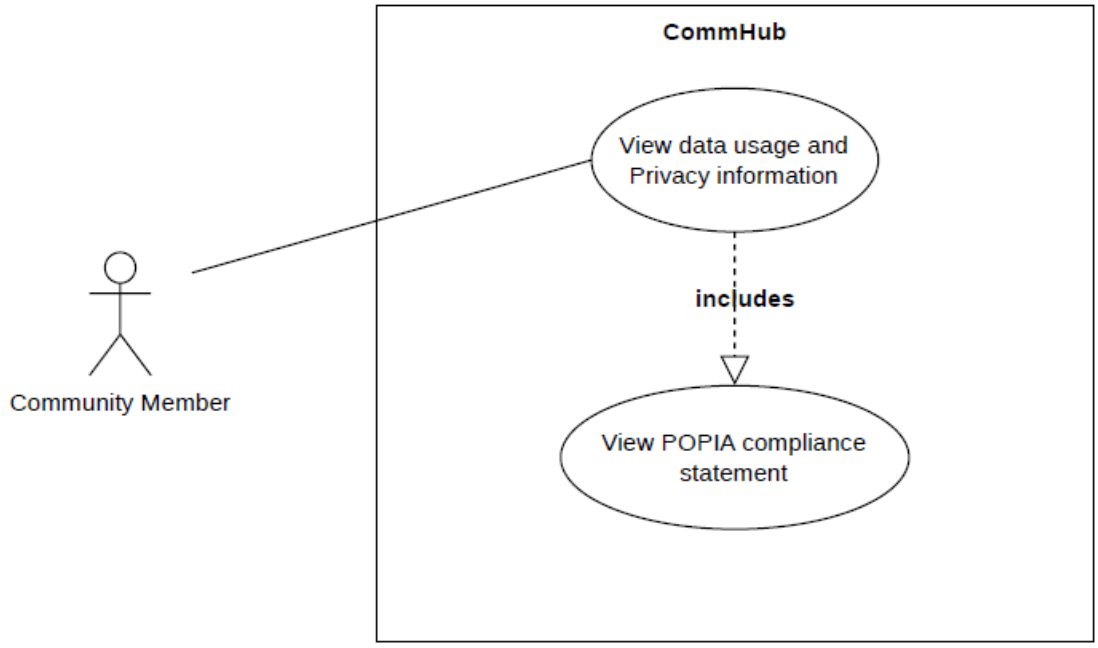
Use Case Diagrams


Account Management
Delete account
<<remove>>
User information
<<include>>
Create Account
:User
:Admin
Platform hosting
Make development environment
Establish internet connection
<<include>>
View web app online
Update web app
View web app offline
:Developer
:User

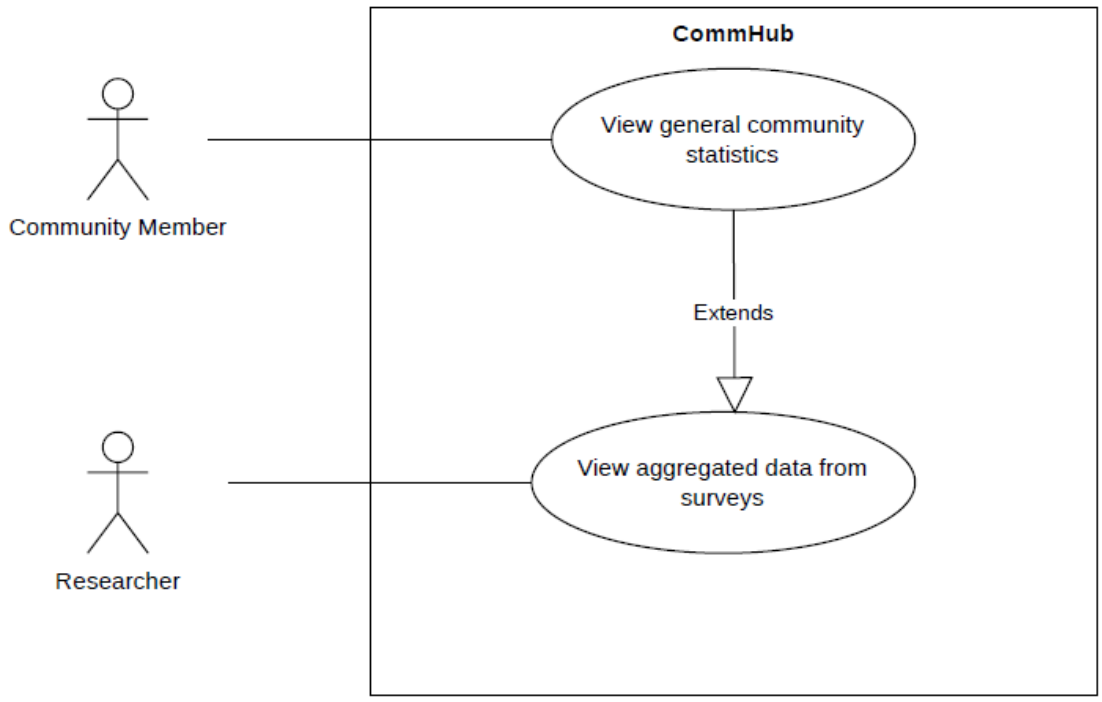

Research Management
Make survery
Publish survey
<<extend>>
Track honorarium payments
<<include>>
Give consent
Fill out survery
Browse surveys
Delete survey
:Researcher
:User

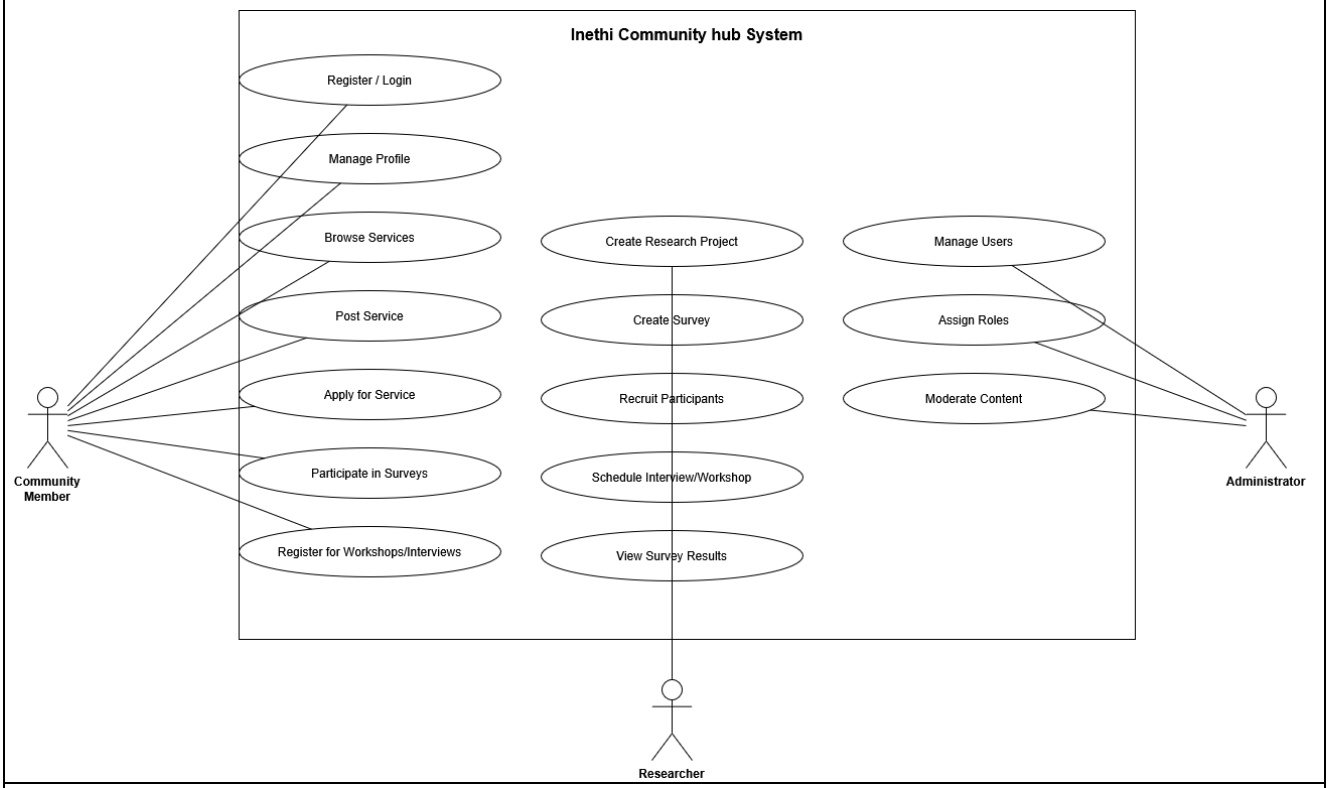

Service Directory
Make Post
Manage applicants
<<include>>
View applicants
Browse posts
Search/filter posts
Delete post
:User

Backlog Item: Trust



Backlog Item: Community Statistics




Inethi Community hub System
Register / Login
Manage Profile
Browse Services
Post Service
Apply for Service
Participate in Surveys
Register for Workshops/Interviews
Create Research Project
Create Survey
Recruit Participants
Schedule Interview/Workshop
View Survey Results
Manage Users
Assign Roles
Moderate Content
Community Member
Administrator
Researcher